

nRF54L Series Production Programming

Application Note

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Revision history

Date	Description
June 2026	Updated: • Selecting the access port on page 6 • Programming on page 11 • Writing data with access port protection disabled on page 11 • Provisioning KMU data on page 12 • Enabling device protection on page 13
November 2024	First release

1 Introduction

This document provides information on writing software to the nRF54L Series devices and is intended for developers of *Non-volatile Memory (NVM)* programming tools.

It serves as a starting point for the nRF54L Series device support in production tools and accelerates the engineering process of supporting the nRF54L Series devices. This document describes a robust method for programming devices. You might not need to follow every step in some cases, for example, if the device has never been programmed before and its NVM is completely erased or if the device is unprotected.

2 Programming flow

The diagram shows the flow of production programming under normal circumstances.

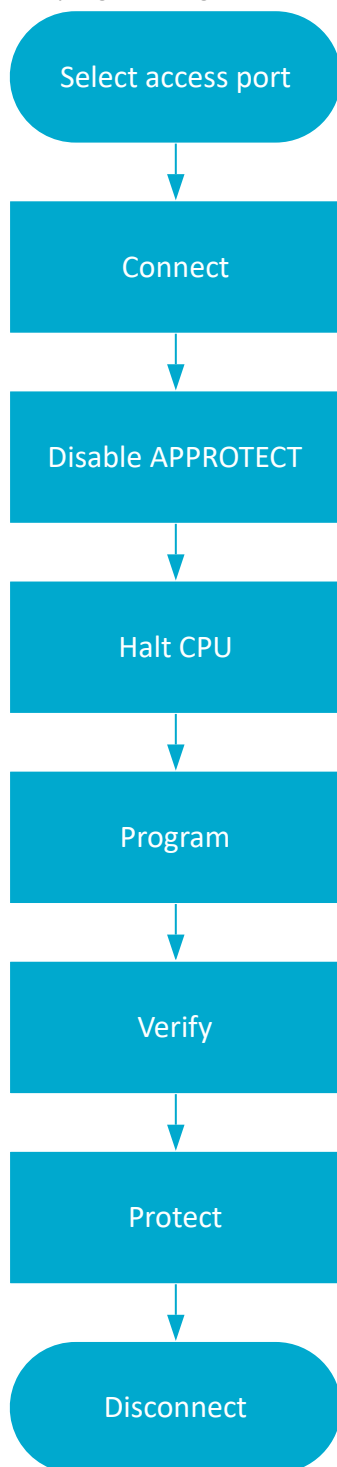


Figure 1: Normal programming flow

3 Selecting the access port

The nRF54L Series devices have one Arm[®] Cortex[®]-M33 processor, and some devices have also a RISC-V coprocessor.

The *NVM* is shared between the present CPUs and programmed using the access port of the Arm Cortex-M33.

When communicating with the device using the *Serial Wire Debug Port (SW-DP)*, the Cortex-M33 core at AHB-AP=0 is selected by default. The control access port uses AHB-AP=2. The coprocessor core, or AUX access port, at AHB-AP=1 is available only for debugging the coprocessor code and cannot be used to program the device. To select the target processor, set AHB-AP to the corresponding value.

4 Connecting

Use the standard *Serial Wire Debug (SWD)* Arm CoreSight™ *Debug Access Port (DAP)* protocol to enter debug interface mode.

Before the external debugger can access the CPU, it must first request the device to power up and make sure that the appropriate power domains are powered up. This is managed using the built-in CxxxPWRUPREQ and CxxxPWRUPACK features found in the DAP. If the debugger requests the debug domain or the complete system to power up, the device stays in debug interface mode.

5 Disabling APPROTECT

If the device has access port protection enabled and is not erase protected, you can disable access port protection by using a *Control Access Port (CTRL-AP)* erase all operation.

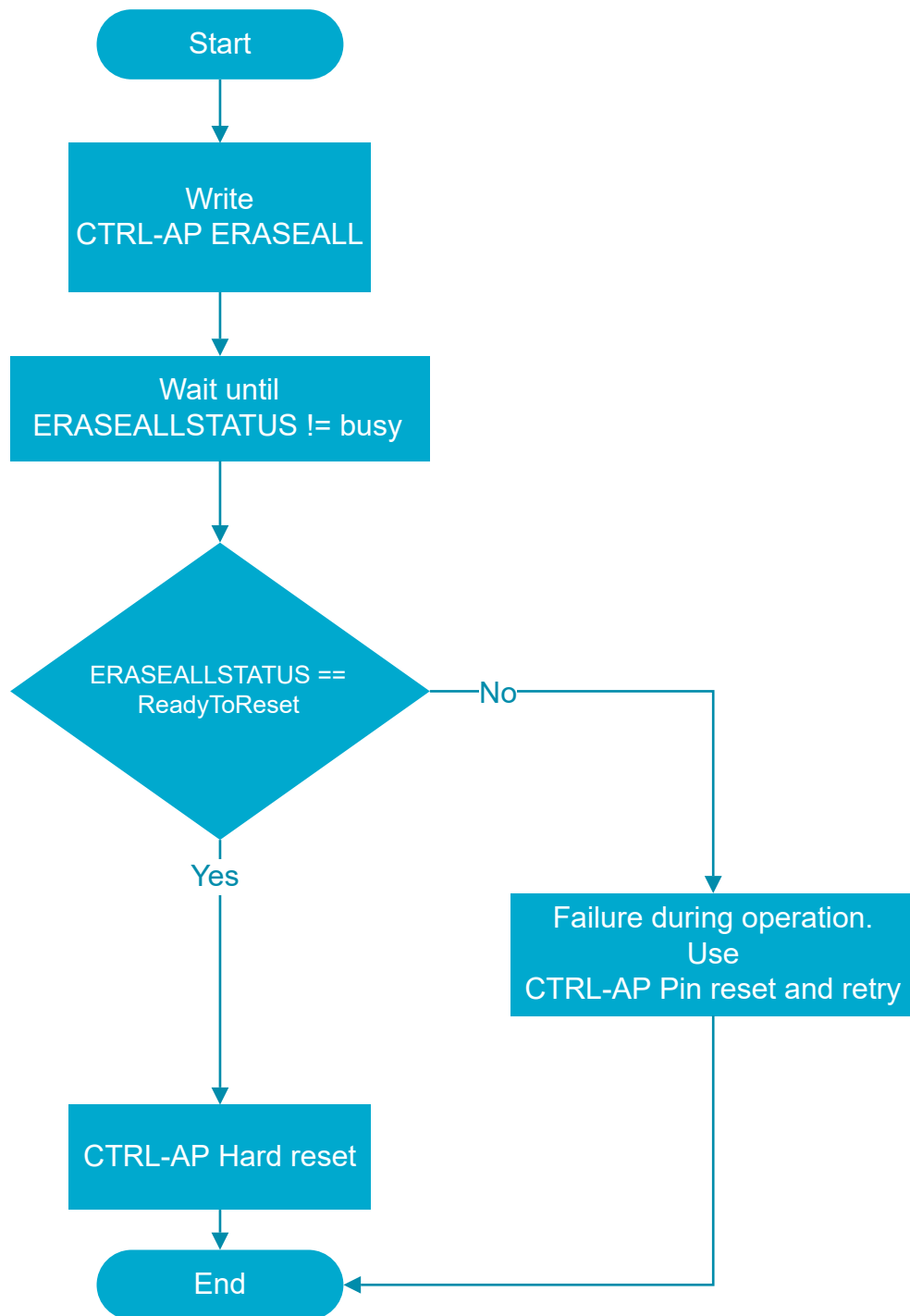
To disable access port protection, complete the following steps:

1. Write 1 (Erase) to the CTRLAP.ERASEALL register (0x004).
2. Wait for the value of the CTRLAP.ERASEALLSTATUS register (0x008) to change from 2 (Busy).
3. Read the value of the CTRLAP.ERASEALLSTATUS register (0x008).
4. If the value of the CTRLAP.ERASEALLSTATUS register (0x008) is 1 (ReadyToReset), complete the following steps to do a hard reset:
 - a) Write 2 (HardReset) to the CTRLAP.RESET register (0x000).
 - b) Write 0 (NoReset) to the CTRLAP.RESET register (0x000).

This completes the operation to disable the access port protection.

If the value of the CTRLAP.ERASEALLSTATUS register (0x008) is a non-zero value other than 1, the operation has failed and a pin reset is needed. To perform a pin reset, write 4 (PinReset) to the CTRLAP.RESET register (0x000), followed by writing 0 (NoReset) to the same register. After the pin reset, return to Step 1 to repeat the procedure.

For more information on erase protection, see [Checking protection status](#) on page 16.

*Figure 2: Erase all states*

6 Halting the CPU

Use the standard *SWD* Arm CoreSight *DAP* protocol to issue a halt command to the chip.

An application running on the device that was previously programmed can use the *Watchdog timer (WDT)*. The default configuration of the WDT pauses it if the CPU is halted.

7 Programming

Programming the device's non-volatile memory (RRAM) and *User Information Configuration Registers (UICR)* is managed through the *Resistive Random-Access Memory Controller (RRAMC)*. In devices that have a key management unit (KMU), the KMU provisions assets into the SICR by copying data and permission policies from RAM to SICR.

When RRAMC write is enabled, you can write to the non-volatile memory or UICR. Only word-aligned writes are allowed when programming the UICR.

7.1 Writing data with access port protection disabled

Use the standard *SWD Arm CoreSight DAP* protocol to write data into the non-volatile memory (RRAM).

Different nRF54L Series devices have different base address for the *RRAMC*. The following table lists the addresses for the different devices.

Register name	nRF54L05, nRF54L10, nRF54L15	nRF54LM20A, nRF54LM20B, nRF54LV10A	nRF54LS05A, nRF54LS05B
RRAMC.TASKS_COMMITWRITEBUF	0x5004B008	0x5004E008	0x4004E008
RRAMC.CONFIG	0x5004B500	0x5004E500	0x4004E500
RRAMC.READY	0x5004B400	0x5004E400	0x4004E400

Table 1: Register overview

The RRAM controller has different write buffer sizes. The following table lists the maximum write buffer size, the corresponding data chunk size, and the corresponding RRAMC.CONFIG register value for the configuration.

Device	Write buffer size in bytes	RRAM data unit size in bytes	Chunk size in bytes	CONFIG register value
nRF54LV10A	16	16	256	0x1001
nRF54L05, nRF54L10, nRF54L15, nRF54LM20A, nRF54LM20B	32	16	512	0x2001
nRF54LS05A, nRF54LS05B	32	8	256	0x2001

Table 2: RRAMC.CONFIG register value to enable write and use maximum buffer size

Complete the following steps:

1. To select the maximum write buffer size for write operations, write the appropriate CONFIG register value from [Table 2: RRAMC.CONFIG register value to enable write and use maximum buffer size](#) on page 11 to the RRAMC.CONFIG register of the RRAMC.
This enables buffered write to the non-volatile memory.
2. Read the RRAMC.READY register until the value is 0x1.

When this value is read, the RRAMC is ready and not performing any operations.

3. Write the data in chunks to the desired, word-aligned address.

The chunk size must match the value specified in [Table 2: RRAMC.CONFIG register value to enable write and use maximum buffer size](#) on page 11. If the data is not aligned to the chunk size, use the RRAMC.TASKS_COMMITWRITEBUF register to commit the data from the internal write buffer to the non-volatile memory. Write 0x1 to the RRAMC.TASKS_COMMITWRITEBUF register to trigger this task.

4. When all data is written, read the RRAMC.READY register again until the value is 0x1 to ensure the write operation has completed.
5. Write 0x0 to the RRAMC.CONFIG register of the RRAMC.
This configures the non-volatile memory as read-only.

The ranges of writeable addresses are the following:

- UICR addresses, located in addresses 0x00FFD000 through 0x00FFDFFC
- All program memory, located in addresses 0x00000000 through ((FICR.INFO.RRAM * 1024) – 4)

7.2 Verifying non-volatile memory

To verify the contents of the non-volatile memory (RRAM) after programming, use the standard SWD Arm CoreSight DAP protocol to read every written address and compare with the expected values.

You can optimize the memory verification process by running a digest calculation of the entire memory on the device and then compare this digest to ensure it has been written correctly.

7.3 Provisioning KMU data

The key management unit (KMU) manages provisioning data and permission policy by copying from RAM to SICR.

Note: This procedure applies only to devices with a KMU. Skip it for nRF54LS05A and nRF54LS05B as they do not have a KMU.

Different nRF54L Series devices have different base address for the KMU and RRAMC. The following table lists the addresses for the different devices.

Register name	nRF54L05, nRF54L10, nRF54L15	nRF54LM20A, nRF54LM20B, nRF54LV10A
KMU.SRC	0x50045504	0x50049504
KMU.KEYSLOT	0x50045500	0x50049500
KMU.TASKS_PROVISION	0x50045000	0x50049000
KMU.EVENTS_PROVISIONED	0x50045100	0x50049100
RRAMC.CONFIG	0x5004B500	0x5004E500
RRAMC.READY	0x5004B400	0x5004E400

Table 3: Register overview

Complete the following steps:

1. Write the KMU SRC data struct as an array to RAM.

2. Write the RAM address of the KMU SRC data to the KMU.SRC register.
3. Write the target key slot ID to the KMU.KEYSLOT register.
4. Set the CONFIG register of the RRAMC to `WEN.Enabled` by writing `0x1` to the RRAMC.CONFIG register.
5. Read the RRAMC.READY register until the value is `0x1`.
6. Trigger KMU provision by writing `0x1` to the KMU.TASKS_PROVISION task.
7. Read the KMU.EVENTS_PROVISIONED event until the value is `0x1`.
8. Clear the event by writing `0x0` to the KMU.EVENTS_PROVISIONED register.
9. Repeat steps 1 to 8 for the required number of key slots to be provisioned.
10. Write `0x0` to the RRAMC.CONFIG register.

The KMU data and permission policies are write-only and can therefore be verified only through a cryptographic operation during production testing.

The KMU data and permission policies can be erased only by using the CTRL-AP erase all operation.

7.4 Enabling device protection

There are several ways to protect the nRF54L Series devices. *Access Port Protection (APPROTECT)* secures the access port, *Erase Protection (ERASEPROTECT)* stops the device from being erased, and *Secure Access Port Protection (SECUREAPPROTECT)* stops unauthorized access to secure resources.

Note: If the device has activated both APPROTECT and ERASEPROTECT, it cannot be recovered without a proper software solution. See [Checking protection status](#) on page 16 for more information.

Devices that feature Arm TrustZone® have individual control over secure and non-secure access port protection.

The following table lists the available protection mechanisms for nRF54L Series devices.

Devices	Available protection mechanisms
nRF54L05, nRF54L10, nRF54L15, nRF54LM20A, nRF54LM20B, nRF54LV10A	SECUREAPPROTECT, APPROTECT, and ERASEPROTECT
nRF54LS05A, nRF54LS05B	APPROTECT and ERASEPROTECT

Table 4: Device protection mechanisms

APPROTECT

APPROTECT locks the *Tamper Controller (TAMPC)* PROTECT.DOMAIN signal protectors, which blocks debugger's read and write access to all CPU registers and memory mapped addresses.

To check if APPROTECT is enabled, read the UICR.APPROTECT[0].PROTECT0 (0x00FFD000) and UICR.APPROTECT[0].PROTECT1 (0x00FFD01C) registers. If these registers have any value other than `0xFFFFFFFF`, the device is protected. If the registers show the device as unprotected, write `0x50FA50FA` to them.

The protection activates after a reset. Ensure all protections are configured before performing a reset if enabling ERASEPROTECT or SECUREAPPROTECT.

ERASEPROTECT

ERASEPROTECT blocks RRAMC ERASEALL and CTRL-AP.ERASEALL functionality.

To check if ERASEPROTECT is enabled, read the UICR.ERASEPROTECT[0].PROTECT0 (0x00FFD060) and UICR.ERASEPROTECT[0].PROTECT1 (0x00FFD07C) registers. If these registers have any value other than 0xFFFFFFFF, the device is protected. If the registers show the device as unprotected, write 0x50FA50FA to them.

The protection activates after a reset. Ensure all protections are configured before performing a reset if enabling APPROTECT or SECUREAPPROTECT.

SECUREAPPROTECT

Note: SECUREAPPROTECT is not available on nRF54LS05A and nRF54LS05B. See [Table 4: Device protection mechanisms](#) on page 13.

SECUREAPPROTECT blocks debugger's read and write access to all secure CPU registers and secures memory mapped addresses.

To check if SECUREAPPROTECT is enabled, read the UICR.SECUREAPPROTECT[0].PROTECT0 (0x00FFD020) and UICR.SECUREAPPROTECT[0].PROTECT1 (0x00FFD03C) registers. If these registers have any value other than 0xFFFFFFFF, the device is protected. If the registers show the device as unprotected, write 0x50FA50FA to it.

The protection activates after a reset. Ensure all protections are configured before performing a reset if enabling APPROTECT or ERASEPROTECT.

Note: A cold boot through a pin reset or power cycling the device activates the TAMPC signal protectors even if the UICR APPROTECT registers are set to unprotected. In this case, protection can be disabled only through an ERASEALL operation. For more information, see [Disabling APPROTECT](#) on page 8.

8 Disconnecting

Use the standard *SWD* Arm CoreSight *DAP* protocol to exit the debug interface mode.

This is managed using the built-in `CxxxPWRUPREQ` and `CxxxPWRUPACK` features found in the Arm CoreSight *DAP*. When the debugger stops requesting the debug domain or the complete system to be powered up, the device exits the debug interface mode.

We recommend a reset of the device that results in cold boot after programming by doing a power cycle.

9 Troubleshooting

The nRF54L Series devices can be reprogrammed by overwriting the memory contents. Reprogramming depends also on the current nRF54L Series security settings.

9.1 Checking protection status

In the nRF54L Series devices, access port protection is enabled by default, but the devices can also have secure access port protection and erase protection.

For more information on access port protection, see [Enabling device protection](#) on page 13.

To check if your device is protected, read the following registers in the AHB-AP:

- AHP-AP Control/Status Word (CSW) register – use this register to read the *APPROTECT* status. This register is defined in the [Arm® CoreSight SoC-400 Technical Reference Manual](#). Use the following fields to check the access port protection status:
 - DgbStatus field (bit 6 in AHB-AP.CSW) – indicates if AHB transfers are permitted. If the value of AHB-AP.CSW->DgbStatus is 1, then AHB transfers are allowed, and the device does not have APPROTECT.
 - SPIStatus field (bit 23 in AHB-AP.CSW) – indicates if secure protection is enabled. If the value of AHB-AP.CSW->SPIStatus is 1, *SECUREAPPROTECT* is not set and both secure and non-secure transfers are allowed.
- CTRL-AP.ERASEPROTECT.STATUS – If access port protection is not enabled, use this register to check if *ERASEPROTECT* is set. If both APPROTECT and ERASEPROTECT are set, the device cannot be unlocked unless programmed software changes the settings.

Glossary

Access Port Protection (APPROTECT)

A register used to prevent read and write access to all CPU registers and memory-mapped addresses.

Control Access Port (CTRL-AP)

A custom access port that enables control of the device even if other access ports in the debug access port are disabled by the access port protection.

Debug Access Port (DAP)

Provides multiple master driving ports, all accessible and controlled through a single external interface port to provide system-wide debug.

Erase Protection (ERASEPROTECT)

A register used to block NVMC ERASEALL, RRAMC ERASEALL, and CTRL-AP.ERASEALL functionality.

Factory Information Configuration Registers (FICR)

Pre-programmed registers that contain chip-specific information and configuration. FICRs cannot be erased by users.

Non-volatile Memory (NVM)

Memory that can retrieve stored information even after having been power-cycled.

Non-volatile Memory Controller (NVMC)

A controller used for writing and erasing the internal flash memory and the *UICR*.

Resistive Random-Access Memory Controller (RRAMC)

A controller used for writing and erasing the non-volatile memory and the *UICR*.

Secure Access Port Protection (SECUREAPPROTECT)

A register used to prevent read and write access to all secure CPU registers and secure memory-mapped addresses.

Serial Wire Debug (SWD)

A standard two-wire interface for programming and debugging Arm CPUs.

Serial Wire Debug Port (SW-DP)

An interface that provides a low pin count bi-directional connection to the DAP with a reference clock signal for synchronous operation.

System on Chip (SoC)

A microchip that integrates all the necessary electronic circuits and components of a computer or other electronic systems on a single integrated circuit.

Tamper Controller (TAMPC)

A controller that manages inputs from internal and external physical attack detectors and controls the device's response.

User Information Configuration Registers (UICR)

Non-volatile memory registers used to configure user-specific settings.

Watchdog timer (WDT)

A timer that causes a system reset if it is not poked periodically.

Recommended reading

In addition to the information in this document, you might need to consult other documents.

Nordic documentation

- [nRF54L15 | nRF54L10 | nRF54L05 Datasheet](#)
- [nRF54LM20A | nRF54LM20B Datasheet](#)
- [nRF54LS05A | nRF54LS05B Datasheet](#)
- [nRF54LV10A Preliminary Datasheet](#)

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